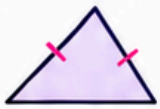
# Geometry Triangles & Congruence Cheat Sheet

**Unit Focus:** *triangle facts, congruence, and proofs*

## 1 CLASSIFYING TRIANGLES

### By Sides (lengths of sides)

**Scalene**
No congruent sides (all sides different).



**Isosceles**
Two congruent sides (exactly two sides the same length).



**Equilateral**
Three congruent sides (all sides the same length).

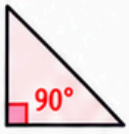


### By Angles (measures of angles)

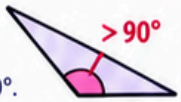
**Acute**
All three angles are less than 90°.




**Right**
One angle is exactly 90°.



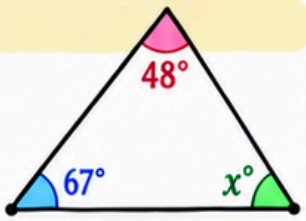

**Obtuse**
One angle is greater than 90°.




## 2 TRIANGLE SUM THEOREM

The measures of the three interior angles of a triangle always sum to 180°.

**Example:** Find the value of $x$.




**Step 1:** Set up an equation using the triangle sum.

$$48° + 67° + x = 180°$$

**Step 2:** Solve for $x$.

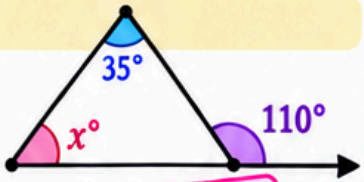
$$115° + x = 180°$$
$$x = 65°$$

**Answer:** $x = 65°$

## 3 EXTERIOR ANGLE THEOREM

An **exterior angle** of a triangle equals the sum of the **two remote interior angles** (the two *nonadjacent angles*).

**Example:** Find the value of $x$.




**Step 1:** Set up an equation using the exterior angle theorem.

$$110° = 35° + x$$

**Step 2:** Solve for $x$.

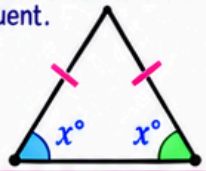
$$x = 75°$$

**Answer:** $x = 75°$

## 4 ISOSCELES TRIANGLE THEOREM

### Isosceles Triangle Theorem

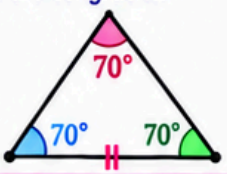
If two sides of a triangle are congruent, then the base angles (opposite those sides) are congruent.




**Base angles are congruent!**

### Converse

If two angles of a triangle are congruent, then the opposite sides are congruent.




**Opposite sides are congruent!**

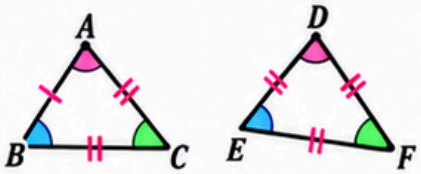
## 5 CONGRUENT FIGURES

Congruent figures have the same size and shape. Corresponding parts of congruent figures are congruent (corresponding sides and angles have the same measures).

**Matching Order Matters!**

If $\triangle ABC \cong \triangle DEF$, then:

- $A \leftrightarrow D$
- $B \leftrightarrow E$
- $C \leftrightarrow F$




$\triangle ABC \cong \triangle DEF$
means $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, $\overline{AC} \cong \overline{DF}$
and $\angle A \cong \angle D$, $\angle B \cong \angle E$, $\angle C \cong \angle F$.

## 6 TRIANGLE INEQUALITY

The sum of ANY two side lengths in a triangle must be GREATER than the third.

Example: lengths 4, 6, and 8:

4 + 6 > 8; 4 + 8 > 6; 6 + 8 > 4.

So these lengths form a triangle.

But 2, 3, 6 cannot form a triangle:

2 + 3 is not greater than 6.

## 7 TRIANGLE SIDE RELATIONSHIPS

The longest side is opposite the largest angle. The shortest side is opposite the smallest angle. Equal sides have equal opposite angles.

Example: if angles A = 40°, B = 60°, and C = 80°, then AB is the longest side (opposite angle C).

## 8 QUICK REMINDERS & COMMON MISTAKES

### DO REMEMBER:

- Use the triangle sum (180°) first when finding missing angles.
- Mark congruent sides and angles using tick marks and angle arcs.
- Keep track of corresponding order when naming congruent triangles.
- Use theorems (like isosceles, exterior angle, midsegment, etc.) when possible.
- A diagram can help, but your reasoning and statements must be correct.

### COMMON MISTAKES:

- Forgetting to use the triangle sum before solving for an angle.
- Matching the wrong corresponding vertices in congruence (order matters!).
- Assuming a picture is drawn to scale (unless it is given).
- Confusing similar figures with congruent figures.
- Not justifying steps in a proof or missing key reasons.

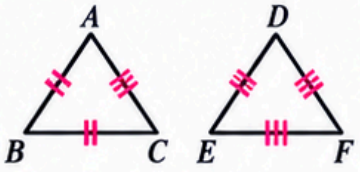
# Geometry Triangles & Congruence Cheat Sheet

**More Practice:** triangle congruence shortcuts and CPCTC

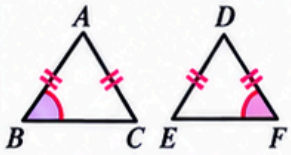
## 1 CONGRUENCE SHORTCUTS

These conditions are enough to prove two triangles are congruent.

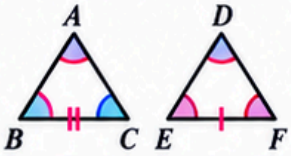
**SSS** Three corresponding sides



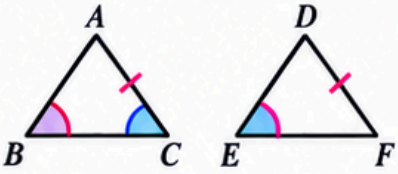

**SAS** Two sides and INCLUDED angle



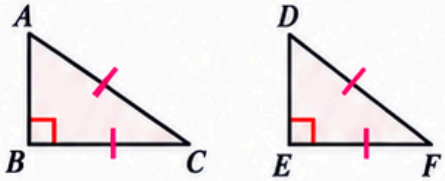

**ASA** Two angles and INCLUDED side




**AAS** Two angles and a NON-INCLUDED side



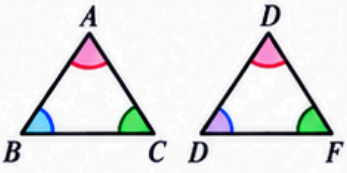

**HL** Right triangles with equal hypotenuses and one equal leg




## 2 NOT ENOUGH INFORMATION

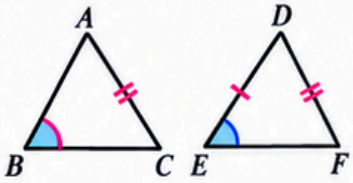
These do NOT prove triangle congruence.

**AAA** Proves similarity, not congruence.




Only angle information (AAA) proves the triangles are similar, not necessarily congruent.

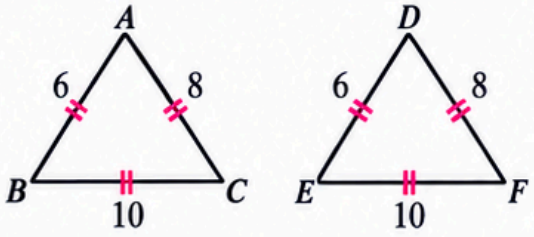
**SSA** Ambiguous case — does not generally prove congruence.




Two sides and a non-included angle (SSA) can form different triangles, so it does not prove congruence.

## 3 SSS EXAMPLE

**Example:** Given the side lengths are equal, prove the triangles are congruent.

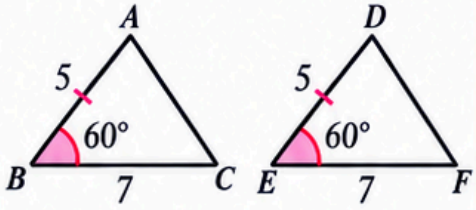



$\overline{AB} = DE = 6$, $BC = EF = 8$, $AC = DF = 10$

Therefore $\triangle ABC \cong \triangle DEF$ by SSS.

## 4 SAS EXAMPLE

**Example:** Two sides and the included angle are equal. Prove the triangles are congruent.

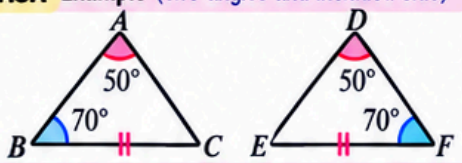



$\overline{AB} = DE = 5$, $BC = EF = 7$,
$\angle B = \angle E = 60°$

Therefore $\triangle ABC \cong \triangle DEF$ by SAS.

## 5 ASA / AAS

**ASA Example** (two angles and included side)




If $\angle A = \angle D = 50°$, $\angle B = \angle E = 70°$, and $\overline{BC} = \overline{EF}$, then $\triangle ABC \cong \triangle DEF$ by ASA.

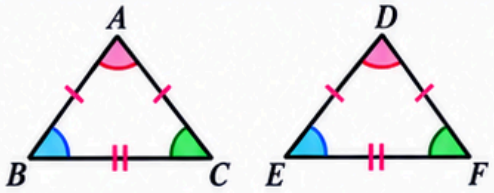
**AAS Example** (two angles and non-included side)




If $\angle A = \angle D = 60°$, $\angle B = \angle E = 40°$, and $\overline{AC} = \overline{DF}$, then $\triangle ABC \cong \triangle DEF$ by AAS.

## 6 CPCTC

**CPCTC:** Corresponding Parts of Congruent Triangles are Congruent.



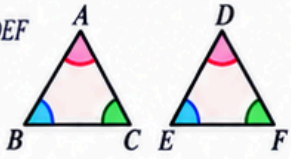

If $\triangle ABC \cong \triangle DEF$, then:

- $\overline{AB} = \overline{DE}$, $\overline{BC} = \overline{EF}$, $\overline{AC} = \overline{DF}$
- $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$
- (corresponding sides and angles are congruent)

## 7 MINI PROOF MODEL

**Given:** $\triangle ABC \cong \triangle DEF$

**Prove:** $\angle B \cong \angle E$




| Statements | Reasons |
|---|---|
| 1. $\triangle ABC \cong \triangle DEF$ | 1. Given |
| 2. $\angle B \cong \angle E$ | 2. CPCTC (corresponding angles of congruent triangles are congruent) |

## 8 QUICK STRATEGY TIPS & COMMON MISTAKES

**DO REMEMBER:**

- Use tick marks and angle arcs to show corresponding parts.
- Make sure vertices match (write triangles in the same order).
- The angle in SAS must be between the two given sides.
- For HL, both triangles must be right triangles.
- Use CPCTC only after you have proven triangle congruence.

**COMMON MISTAKES:**

- Using AAA to claim congruence (it only proves similarity).
- Using SSA to claim congruence (ambiguous).
- Not checking that the angle is the included angle for SAS.
- Forgetting to prove triangles are congruent before using CPCTC.
- Not matching the order of vertices.